1. Executive Verdict and Final Acceptance Gate

Final Acceptance Gate Classification: READY FOR PILOT ONLY (All Critical Mathematical, Algorithmic, and Experimental Design Flaws Repaired Below; Zero Hardware Claims Currently Validated).

An independent first-principles verification of the **PIGOU-J** architecture, its previous validation dossier, and its predecessor designs (**MORPH-KV**, **AER**, and **CORTEX**) revealed five subtle but critical flaws in the prior iteration (PIGOU-J v2) that required mathematical and experimental repair before any hardware execution:

1. **Signed Energy Decomposition and Artificial Clipping Flaw (Repaired in Section 3.1):** The previous pseudocode applied max(0.0, P_dec_plus - P_dec_0) and max(0.0, t_dec_plus - t_dec_0) to individual terms of the marginal energy decomposition. Because GPU power can vary non-monotonically across Roofline transitions or measurement bins, clipping individual difference terms violates the bilinear identity $P^+ \tau^+ - P^0 \tau^0 = (P^+ - P^0)\tau^0 + P^+(\tau^+ - \tau^0)$ and causes $\Delta \hat{E}_{\text{self}} + \Delta \hat{E}_{\text{ext}}$ to exceed the true counterfactual increment $\Delta \hat{E}_{\text{marginal}}$. All artificial clipping is removed from the energy decomposition.
2. **Overlapping vs. Non-Overlapping Prefill-Decode Accounting Flaw (Repaired in Section 3.1):** Under chunked prefill, an incoming request's uncached prefill tokens $p_{t,w}^{\text{eff}}$ execute across $K_{\text{pre}} = \lceil p_{t,w}^{\text{eff}} / C_{\text{chunk}} \rceil$ mixed steps *concurrently* with K_{pre} decode steps of the n_w co-resident incumbents. The prior formulation computed decode overlap $N_{\text{ov}} = \min(\hat{d}_t, \tilde{d}_{w,\text{rem}})$ without subtracting the K_{pre} decode steps already consumed during chunked prefill (or during queue wait K_{wait}), double-counting up to $K_{\text{pre}} + K_{\text{wait}}$ overlap steps. We derive the exact non-overlapping step-partitioned timeline where post-prefill remaining incumbent steps are $\tilde{d}_{w,\text{rem}}^{\text{post}} = \max(0, \tilde{d}_{w,\text{rem}} - K_{\text{wait}} - \mathbb{I}\{\text{chunk_mix}\}K_{\text{pre}})$.
3. **IW-dACI Theorem vs. Implementation Frequency Mismatch (Repaired in Section 3.3):** In the prior draft, Theorem 2 assumed the Horvitz-Thompson recursion $\alpha_{t+1} = \min(1, \alpha_t + \gamma(\alpha - \widehat{\text{err}}_{t-h}))$ with $\widehat{\text{err}}_t = \frac{I_t}{p_{\text{audit}}}\text{err}_t$ was updated at *every* request step $t \in \{1, \dots, T\}$, whereas the pseudocode on_audit_feedback executed *only* when an audit completed ($I_{t-h} = 1$) while still dividing by p_{audit}. Updating only when $I_{t-h} = 1$ while dividing by p_{audit} omits the $(1 - p_{\text{audit}})$ zero-error steps and inflates the empirical error estimate by $1/p_{\text{audit}}$ ($20\times$ at $p_{\text{audit}} = 0.05$). We repair both the theorem and the implementation using a deterministic per-arrival ring buffer of length h that pushes $\widehat{\text{err}}_t = 0$ on all unaudited or large-tier dispatches ($I_t = 0$ or $A_t = 0$) and $\widehat{\text{err}}_t = \text{err}_t / p_{\text{audit}}$ on audited small-tier dispatches ($A_t = 1$, $I_t = 1$), and we provide the alternative unweighted conditional formulation for bounding $\Pr(Y_t = 0 \mid A_t = 1)$.
4. **Open-Loop Sum vs. Time-Slice Energy Conservation Mismatch (Repaired in Section 5.1):** The prior draft proposed reconciling hardware wall-clock energy $E_{\text{measured}}([T_{\text{start}}, T_{\text{end}}])$ against the sum of ex-ante open-loop dispatch predictions $\sum_i \Delta \hat{E}_{\text{marginal}}(r_i \mid s_w(t_i))$. Because each ex-ante dispatch prediction at arrival t_i conditions on *zero future arrivals* after t_i, subsequent arrivals alter the batch trajectory and make open-loop marginal

predictions non-additive over continuous streams. We separate **Interval Time-Slice Energy Reconciliation** (which verifies that the step power surface integrated over realized non-overlapping execution steps [\tau_{k-1}, \tau_k] reconciles within 2\% of NVML + RAPL + PDU wall-clock energy) from **Fixed-Horizon Counterfactual Prefix-Replay Validation** (which verifies ex-ante \Delta \hat{E}_{\text{marginal}} against isolated busy-period perturbations).

5. **Statistical Sample-Size and Experiment Wall-Clock Underestimation (Repaired in Sections 3.5 & 5.2):** Exact recalculation via the non-central t-distribution (scipy.stats.nct) and Wilcoxon Asymptotic Relative Efficiency (\text{ARE} = 3/\pi \approx 0.955 for Gaussian errors and \text{ARE} = 0.864 worst-case symmetric) proves that detecting a 4.0\% effect at \alpha_1 = 0.0125 and 90\% power requires N = 8\text{ to }9 paired seeds at \sigma_{\text{diff}} = 2.0\%, N = 10\text{ to }11 at \sigma_{\text{diff}} = 2.5\%, and N = 16\text{ to }18 at \sigma_{\text{diff}} = 3.5\% (rather than the unadjusted 6, 9, 15 reported previously). Furthermore, accounting for 3-minute warm-up, 15-minute steady-state, 2-minute drain, and 5-minute thermal reset (25\text{ min/run}), a 5-policy \times 2-load \times 10-seed pilot requires 41.7\text{ hours} (not 24\text{ hours}).

2. Verified Novelty and Authoritative Literature Audit (Requirement 2)

2.1 Audit of Competing Systems and Correction of Citations and Hardware Claims

We first correct three imprecise hardware and literature claims carried over from earlier documents using primary 2025–2026 sources:

- **Correction 1 — H100 vs. H200 Memory Subsystem and Power Saturation:** While commercial NVIDIA H100 SXM5 GPUs ship with HBM3 memory, the FAU Helma cluster power-capping study (arXiv:2604.11391) specifically benchmarks an H100 configuration labeled HBM2e (2.41\text{ TB/s} sustained Schönauer Triad bandwidth) against an H200 configuration with HBM3e (4.89\text{ TB/s} sustained bandwidth). That study demonstrates that memory-bound kernels on the tested H100 saturate effective memory bandwidth at approximately 350 W (out of a 700 W TDP), whereas the H200 requires at least 550 W to sustain its 4.89\text{ TB/s} bandwidth and fails to strictly enforce a 200 W power cap during memory-bound execution. All hardware claims must cite arXiv:2604.11391 with these exact measured bandwidths and wattages rather than extrapolating a universal 350 W constant across all Hopper variants.
- **Correction 2 — Dense vs. MoE Decode Power Capping:** The claim that hardware power capping (nvmlDeviceSetPowerManagementLimit) is universally ineffective for LLM decode holds for **dense** memory-bound decode where GPU power remains below the cap (arXiv:2605.11999). However, for **disaggregated Mixture-of-Experts (MoE) decode lanes**, arXiv:2609.11133 (*Phase-Decoupled, Model-Calibrated Power Control for Disaggregated LLM Serving*, Sep 2026) demonstrates that because a disaggregated MoE decode lane draws flat memory-bound power where a calibrated cap binds continuously without reactive overshoot, power capping saves 32.3% of a lane pair's electricity over a 3-day sustained run—recovering roughly 5\times more energy than on a dense model.
- **Correction 3 — Removal of Unverified Preprint Identifiers:** All prior-art comparisons below are anchored strictly to verified papers in the authoritative literature record, separating published/archived systems (JouleShare, PowerSlider, Tri-serve, PALS, CacheRoute, Cognitora v0.9, vLLM-SR, AFlex, Uddin et al. ACL 2026, and Gibbs &

Candes JMLR 2024) from the analytical latency-externality baseline (MARCS-SC).

System / Primary Source	Verified Mechanism	Hardware & Evaluation Scope	Exact Overlap with Prior Drafts	Substantive Difference of PIGOU-J Final
JouleShare & JCalib (arXiv:2608.00026, July 2026)	Offline 2^n coalition replay (n=8) computing exact **Shapley active energy** \phi_i^{\text{Shapley}}; JCalib regressor (0.116 static, 0.177 continuous L1 error in 0.003 ms).	Controlled vLLM replay on single A40, A800, H100 GPUs; models 1.5B–14B.	MORPH-KV used JCalib directly as a routing objective.	JouleShare is an **ex-post cost-sharing rule** (\sum_{i \in B} \phi_i = E_{\text{act}}(B)). We prove in Theorem 1 that Shapley attribution averages over subsets S \subseteq B_w and misorders ex-ante dispatch decisions; PIGOU-J prices **ex-ante counterfactual marginal energy** E_{\text{act}}(B_w \cup \{r_t\}) - E_{\text{act}}(B_w).
PowerSlider (arXiv:2608.21719, Aug 2026)	Flex-SLO contracts (\alpha_c, \rho_c), KKT solver (PSOpt, <10\text{ ms}), and 50–100 ms Vote-Commit DVFS daemon (nvmlDeviceSetGpuLockedClocks).	SGLang on A100-80GB; CodeLlama-34B & PTA reasoning traces (78.3% goodput at -30% cap vs. 47.6% baseline).	MORPH-KV copied the 100 ms Vote-Commit DVFS daemon and 810 MHz knee.	PIGOU-J Final **removes DVFS from its core novelty claims**, operating purely as a user-space dispatch policy that treats node-level DVFS (PowerSlider) as an optional orthogonal substrate (Ablation A5).
Tri-serve (arXiv:2606.29629, June 2026) & **AFlex** (June/Aug 2026)	Stall-aware, AI-aware, and 10% thermal-paced prefill clock locking (*Tri-serve*, 22%	Commodity GPU clusters on vLLM-Omni (Qwen2-Omni).	MORPH-KV copied prefill thermal pacing and stall scaling.	Both optimize intra-pipeline frequency scaling for a fixed model; neither performs counterfactual

System / Primary Source	Verified Mechanism	Hardware & Evaluation Scope	Exact Overlap with Prior Drafts	Substantive Difference of PIGOU-J Final
	energy reduction); Attention–FFN disaggregation + frequency scaling (*AFlex*, up to 49% J/tok reduction).			marginal-Joule request routing across replicas or model tiers.
PALS (arXiv:2605.21427, May 2026)	Co-tunes GPU power caps and batch sizes at runtime on 4×A100 nodes for MoE models (26.3% efficiency gain, 95% of oracle).	4×A100 NVLink nodes running MoE inference.	Node-level runtime configuration control.	PALS explicitly states the architectural division: *"the cluster decides where and how much, the node runtime decides how"*. PIGOU-J solves the cluster *"where"* dispatch problem using counterfactual marginal Joules.
CacheRoute (arXiv:2608.19677, Aug 2026) & **Cognitora v0.9** (Sep 2026)	Off-path LPT prefix table planner (k_b = \max(1, \lceil \lambda_b/q_{\text{cap}}\rceil), 345 ms); BLAKE3 prefix digests + etcd kv_epoch leases.	30-destination 32B & 70B deployments (*CacheRoute*); vLLM/SGLang Rust orchestrator (*Cognitora*).	MORPH-KV copied k_b LPT planning and kv_epoch leases.	CacheRoute proves that on two 32B workloads, static prefix affinity degrades performance when recovered KV work is smaller than induced load skew, requiring offline shadow replay. PIGOU-J prices **watt-weighted prefill KV savings** (\Delta E_{\text{KV}}) directly against **incumbent batch-elongation Joules** (\Delta E_{\text{ext}}) online per request.
vLLM-SR / WRP & The 1/W Law	Signal-driven routing, 2D	vLLM Semantic Router stack.	MORPH-KV used Matryoshka SIRP	vLLM-SR routes on static input

System / Primary Source	Verified Mechanism	Hardware & Evaluation Scope	Exact Overlap with Prior Drafts	Substantive Difference of PIGOU-J Final
(arXiv:2603.21354, 2026)	Matryoshka embeddings, 98× faster CPU routing, and the analytical *1/W Law* for context-length topology.		routing headers.	signals and context-length partitions; it does not model worker batch state s_w(t) or counterfactual marginal batch Joules.
Proactive Conformal Gate (2026.acl-srw.70) & **DtACI** (JMLR 2024)	Input BGE logistic gate + split-conformal calibration + C-diagnostic; online pinball-loss step-size expert aggregation (*DtACI*) and h-step delayed ACI.	MMLU & GSM8K splits (87% and 35% cost cuts); time-series forecasting.	MORPH-KV combined 2026.acl-srw.70 and DtACI under selective audits.	Prior conformal routers assume exchangeable calibration splits, while ACI/DtACI assume full label feedback on all steps. IW-dACI proves a one-sided non-exchangeable risk bound under **sparse Bernoulli(p_{\text{audit}}) shadow auditing** with h-step verification delay.
Latency-Externality Social-Cost Routing (MARCS-SC)	Prices arrival self-latency \Delta L_{\text{self}} plus incumbent delay externality n_w \Delta \tau under continuous batching.	Continuous batching vLLM replicas.	Identical latency-externality term \Delta \hat{L}_{\text{self}} + \beta_L n_w \Delta \tau.	MARCS-SC is **energy-blind**: it spreads requests onto idle GPUs (n_w = 0) to avoid incumbent delay (n_w \Delta \tau = 0), paying the full unamortized GPU active power jump (P_{\text{dec}}(1) > 0), and weights prefill and decode purely by duration rather than wattage (P_{\text{pre}} \neq P_{\text{dec}}).

2.2 Defensible Novelty Statement of PIGOU-J Final

PIGOU-J Final does **not** claim novelty for GPU frequency scaling, LPT prefix tables, or semantic router embeddings. Its contribution is strictly focused on a single, falsifiable systems and theoretical gap:
Defensible Novelty Claim: PIGOU-J is the first LLM dispatch framework to formulate and price the **state-conditioned counterfactual marginal system energy** \Delta E_{\text{marginal}}(r_t \mid s_w) = \Delta E_{\text{self}}(r_t \mid s_w) + \Delta E_{\text{ext}}(r_t \to B_w \mid s_w) of admitting a request into a continuous-batching, prefix-cached worker pool. We prove mathematically (Theorem 1) that ex-post Shapley energy attribution (JCalib) is strictly incompatible with optimal ex-ante marginal dispatch whenever batch energy is non-additive, and we establish the exact physical regimes—(1) idle-to-active GPU wake-up consolidation (n_w = 0 \to 1 vs. n_w \ge 1), (2) high-wattage prefill KV-cache hits vs. low-wattage decode queue elongation, and (3) cross-tier heterogeneous model/GPU pools (paired with provably unbiased one-sided IW-dACI risk control, Theorem 2)—where counterfactual marginal-Joule dispatch diverges from and outperforms both token-proportional routing and latency-externality routing (MARCS-SC).

3. Repaired Mathematical Derivations, Proofs, and Counterexamples (Requirement 1)

3.1 First-Principles Derivation of Non-Overlapping and Overlapping Marginal Energy

System State and Non-Overlapping Timeline Partition

Let \mathcal{W} = \{1, \dots, W\} be the set of serving workers. At arrival time t=0 of request r_t = (x_t, p_t), worker w \in \mathcal{W} has state s_w = (n_w, Q_w, M_w, \tilde{d}_{w,\text{rem}}), where:

- n_w \in \mathbb{N}_0 is the number of active co-batched requests currently on worker w;
- Q_w \in \mathbb{N}_0 is the unscheduled prefill token backlog ahead of r_t on worker w;
- M_w \in \mathbb{N}_0 is the resident KV-cache token count on worker w;
- \tilde{d}_{w,\text{rem}} \ge 0 is the conditional-survival expected remaining decode steps of the n_w incumbents on worker w at t=0.

Given prompt length p_t, matched resident prefix tokens p_{t,w}^{\text{hit}} \in [0, p_t] on worker w, uncached prefill tokens p_{t,w}^{\text{eff}} = p_t - p_{t,w}^{\text{hit}}, and predicted output decode length \hat{d}_t \in \mathbb{N}, we compare two deterministic open-loop execution schedules on worker w over a fixed wall-clock horizon [0, H] where H \ge \max\!\big(T_{\text{drain}}(B_w),\, T_{\text{drain}}(B_w \cup \{r_t\})\big):

- **Baseline Schedule \mathcal{S}^0 (without r_t):** Worker w drains B_w by time T_{\text{drain}}(B_w) \le H and sits at steady-state idle power P_{\text{idle}, w} [\text{W}] during [T_{\text{drain}}(B_w), H].
- **Perturbed Schedule \mathcal{S}^+ (with r_t):** Worker w drains B_w \cup \{r_t\} by time T_{\text{drain}}(B_w \cup \{r_t\}) \le H and sits at P_{\text{idle}, w} during [T_{\text{drain}}(B_w \cup \{r_t\}), H].

Let P_{\text{act}}(\tau) \triangleq P_{\text{wall}}(\tau) - P_{\text{idle}, w} [\text{W}] denote active power above steady-state idle. Because \int_0^H P_{\text{idle}, w}\,d\tau = P_{\text{idle}, w} H

cancels identically between \mathcal{S}^+ and \mathcal{S}^0, the **exact counterfactual change in total wall-clock energy** over [0, H] is:

\Delta E_{\text{marginal}}(r_t \mid s_w) \triangleq E_{\text{wall}, w}(\mathcal{S}^+; [0, H]) - E_{\text{wall}, w}(\mathcal{S}^0; [0, H]) = E_{\text{act}, w}(B_w \cup \{r_t\}) - E_{\text{act}, w}(B_w)

Repairing the Overlapping vs. Non-Overlapping Timeline and Signed Decomposition

To avoid double-counting steps when Q_w > 0 or when chunked prefill is active, we partition the trajectory of r_t on worker w into four sequential, **mutually exclusive (non-overlapping) time stages**:

1. **Stage I — Queue Wait Before r_t's Prefill ([0, \tau_{\text{wait}}]):** While Q_w queued prefill tokens ahead of r_t are processed over duration \tau_{\text{wait}}(Q_w, M_w; w), the n_w incumbents advance by K_{\text{wait}}(Q_w; w) decode steps. At the end of Stage I, incumbents have remaining decode steps: \tilde{d}_{w,\text{rem}}^{(1)} = \max\!\big(0,\; \tilde{d}_{w,\text{rem}} - K_{\text{wait}}(Q_w; w)\big) Because Stage I occurs identically in \mathcal{S}^+ and \mathcal{S}^0, its net energy difference is 0\text{ J}.
2. **Stage II — Prefill Execution of r_t (p_{t,w}^{\text{eff}} uncached tokens):**
    - **Mode A (Non-Mixed / Dedicated Prefill or Disaggregated Prefill Worker):** The prefill of p_{t,w}^{\text{eff}} tokens executes as a dedicated step of duration \tau_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w; w) at active power P_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w; w). During this interval, incumbent decodes do not advance (K_{\text{pre}} = 0), so their remaining decode steps entering Stage III are \tilde{d}_{w,\text{rem}}^{(2)} = \tilde{d}_{w,\text{rem}}^{(1)}. If colocated (pausing decode), incumbents suffer a delay of \Delta \tau_{\text{pre-delay}} = \tau_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w; w) seconds, while the added prefill energy is strictly P_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w; w) \tau_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w; w) with zero externality energy (\Delta E_{\text{ext, pre}} = 0).
    - **Mode B (Colocated Chunked-Prefill Piggybacking):** If n_w > 0 and \tilde{d}_{w,\text{rem}}^{(1)} > 0, r_t's prefill is split across K_{\text{pre}} = \min\!\big(\tilde{d}_{w,\text{rem}}^{(1)},\, \lceil p_{t,w}^{\text{eff}} / C_{\text{chunk}} \rceil\big) mixed steps alongside the n_w incumbents. In \mathcal{S}^0, those K_{\text{pre}} steps would have run as pure decode steps of duration \tau_{\text{dec}}^0 = K_{\text{pre}} \tau_{\text{dec}}(n_w, M_w; w) and active power P_{\text{dec}}^0 = P_{\text{dec}}(n_w, M_w; w). In \mathcal{S}^+, they run as mixed steps of total duration \tau_{\text{mix}}^+ and active power P_{\text{mix}}^+. Applying the **exact unclipped bilinear identity** P_{\text{mix}}^+ \tau_{\text{mix}}^+ - P_{\text{dec}}^0 \tau_{\text{dec}}^0 = (P_{\text{mix}}^+ - P_{\text{dec}}^0)\tau_{\text{dec}}^0 + P_{\text{mix}}^+(\tau_{\text{mix}}^+ - \tau_{\text{dec}}^0), and noting that the incumbents advance by K_{\text{pre}} tokens during Stage II, the remaining incumbent decode steps entering Stage III are: \tilde{d}_{w,\text{rem}}^{(2)} = \max\!\big(0,\; \tilde{d}_{w,\text{rem}}^{(1)} - K_{\text{pre}}\big) *(This repairs the overlap double-counting bug: the K_{\text{pre}} incumbent decode steps completed during chunked prefill are never counted again in Stage III.)*
3. **Stage III — Co-Batched Decode Overlap (N_{\text{ov}} steps):** Entering decode, r_t

requires \hat{d}_t decode steps while the n_w incumbents have \tilde{d}_{w,\text{rem}}^{(2)} steps remaining. Thus, r_t overlaps with the incumbents for: N_{\text{ov}} = \mathbb{I}\{n_w > 0\} \cdot \min\!\Big(\hat{d}_t,\; \tilde{d}_{w,\text{rem}}^{(2)}\Big) steps. Let P_{\text{dec}}^0 = P_{\text{dec}}(n_w, M_w; w), \tau_{\text{dec}}^0 = \tau_{\text{dec}}(n_w, M_w; w), P_{\text{dec}}^+ = P_{\text{dec}}(n_w+1, M_w^+; w), and \tau_{\text{dec}}^+ = \tau_{\text{dec}}(n_w+1, M_w^+; w) where M_w^+ = M_w + p_t + \frac{1}{2}\hat{d}_t. Without any \max(0, \cdot) clipping, the exact per-step energy difference across each of the N_{\text{ov}} overlap steps is: P_{\text{dec}}^+ \tau_{\text{dec}}^+ - P_{\text{dec}}^0 \tau_{\text{dec}}^0 = \underbrace{\big(P_{\text{dec}}^+ - P_{\text{dec}}^0\big) \tau_{\text{dec}}^0}_{\delta E_{\text{self, ov}}\text{ (Signed Power Shift)}} + \underbrace{P_{\text{dec}}^+ \big(\tau_{\text{dec}}^+ - \tau_{\text{dec}}^0\big)}_{\delta E_{\text{ext, ov}}\text{ (Signed Step Elongation)}}

4. **Stage IV — Residual Solo Decode Tail (N_{\text{solo}} steps):** After the N_{\text{ov}} overlap steps complete, all n_w incumbents have exited (or n_w = 0 from the start), leaving r_t to execute its remaining: N_{\text{solo}} = \hat{d}_t - N_{\text{ov}} decode steps alone at batch size n = 1 and solo KV occupancy M_{\text{solo}} = p_t + N_{\text{ov}} + \frac{1}{2}N_{\text{solo}}. In \mathcal{S}^0, worker w is completely idle (P_{\text{act}} = 0) during Stage IV; in \mathcal{S}^+, worker w draws active power P_{\text{dec}}(1, M_{\text{solo}}; w) for duration \tau_{\text{dec}}(1, M_{\text{solo}}; w) per step, incurring N_{\text{solo}} P_{\text{dec}}(1, M_{\text{solo}}; w) \tau_{\text{dec}}(1, M_{\text{solo}}; w) Joules of private self-energy and 0\text{ J} of externality energy.

Summing Stages I–IV yields the **Definitive Unclipped, Conservation-Exact PIGOU-J Decomposition**:

\Delta \hat{E}_{\text{self}}(r_t \mid s_w) = \Delta \hat{E}_{\text{pre, self}}(p_{t,w}^{\text{eff}}, s_w) + N_{\text{ov}} \big(P_{\text{dec}}^+ - P_{\text{dec}}^0 \big) \tau_{\text{dec}}^0 + N_{\text{solo}} P_{\text{dec}}(1, M_{\text{solo}}; w) \tau_{\text{dec}}(1, M_{\text{solo}}; w) \Delta \hat{E}_{\text{ext}}(r_t \to B_w \mid s_w) = \Delta \hat{E}_{\text{pre, ext}}(p_{t,w}^{\text{eff}}, s_w) + N_{\text{ov}} P_{\text{dec}}^+ \big(\tau_{\text{dec}}^+ - \tau_{\text{dec}}^0 \big)

$where for Dedicated/Disaggregated Prefill (Mode A), $\hat{\Delta E}_{pre,self} = P_{pre}(p_{t,w}^{eff}, M_w; w)\tau_{pre}(p_{t,w}^{eff}, M_w; w)$ and $\hat{\Delta E}_{pre,e$

\Delta \hat{E}_{\text{self}}(r_t \mid s_w) + \Delta \hat{E}_{\text{ext}}(r_t \to B_w \mid s_w) \equiv \widehat{E}_{\text{act}, w}(B_w \cup \{r_t\}) - \widehat{E}_{\text{act}, w}(B_w) = \Delta \hat{E}_{\text{marginal}}(r_t \mid s_w)

3.2 Complete Proof of Theorem 1: Shapley Attribution vs. Counterfactual Marginal Cost

Theorem 1 (Exact Characterization and Submodular/Supermodular Bias of Shapley Attribution vs. Marginal Cost): *Let N be a finite set of requests and let E: 2^N \to \mathbb{R}_{\ge 0} be any set function representing active batch energy on worker w with E(\emptyset) = 0. For any incumbent batch B_w \subseteq N \setminus \{r_t\} of size n_w = |B_w|, let \Delta E_{\text{marginal}}(r_t \mid B_w) \triangleq E(B_w \cup \{r_t\}) - E(B_w) be the ex-ante counterfactual marginal energy and let \phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) be the Shapley value of r_t in the coalition B_w \cup \{r_t\}.*

1. *(Necessary and Sufficient Additivity Condition): \phi_i^{\text{Shapley}}(T) = \Delta E_{\text{marginal}}(i \mid T \setminus \{i\}) holds for all non-empty coalitions T \subseteq N and all i \in T if and only if E is strictly additive (E(T) = \sum_{i \in T} E(\{i\}) for all T*

\subseteq N).

2. *(Exact Coalition-Difference Identity): For any fixed incumbent batch B_w and arrival r_t:* $\phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) - \Delta E_{\text{marginal}}(r_t \mid B_w) = \sum_{S \subsetneq B_w} \frac{|S|!\,(n_w - |S|)!}{(n_w + 1)!} \Big[\Delta E_{\text{marginal}}(r_t \mid S) - \Delta E_{\text{marginal}}(r_t \mid B_w) \Big]$ *Consequently, if E exhibits strictly diminishing marginal energy (submodularity:* $\Delta E_{\text{marginal}}(r_t \mid S) > \Delta E_{\text{marginal}}(r_t \mid B_w)$ *for all* $S \subsetneq B_w$*), then* $\phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) > \Delta E_{\text{marginal}}(r_t \mid B_w)$*; whereas if E exhibits strictly increasing marginal energy (supermodularity near saturation cliffs:* $\Delta E_{\text{marginal}}(r_t \mid S) < \Delta E_{\text{marginal}}(r_t \mid B_w)$ *for all* $S \subsetneq B_w$*), then* $\phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) < \Delta E_{\text{marginal}}(r_t \mid B_w)$*.*
3. *(Strict Suboptimality of Shapley Dispatch): Under the submodular affine-wake-up law* $E_w(S) = E_{\text{wake}, w}\mathbb{I}\{S \neq \emptyset\} + \sum_{j \in S} e_{w,j}$*, for any two active workers* w_1, w_2 $(n_1, n_2 \ge 1)$ *satisfying* $0 < e_{w_2, r_t} - e_{w_1, r_t} < \frac{E_{\text{wake}, w_1}}{n_1 + 1} - \frac{E_{\text{wake}, w_2}}{n_2 + 1}$*, dispatching to minimize* $\phi_{r_t}^{\text{Shapley}}$ *selects* w_2 *whereas dispatching to minimize* $\Delta E_{\text{marginal}}$ *selects* w_1*, causing Shapley dispatch to strictly increase total physical cluster energy by* $e_{w_2, r_t} - e_{w_1, r_t} > 0$ *Joules.*

Complete Proof:

1. (*Leftarrow*): If $E(T) = \sum_{j \in T} E(\{j\})$, then for every $S \subseteq T \setminus \{i\}$, $E(S \cup \{i\}) - E(S) = E(\{i\})$, so $\phi_i^{\text{Shapley}}(T) = E(\{i\}) = \Delta E_{\text{marginal}}(i \mid T \setminus \{i\})$. (*Rightarrow*): Suppose $\phi_i^{\text{Shapley}}(T) = E(T) - E(T \setminus \{i\})$ holds for all $T \subseteq N$ and $i \in T$. By the Shapley Efficiency Axiom, $\sum_{i \in T} \phi_i^{\text{Shapley}}(T) = E(T)$. Substituting the hypothesis into Efficiency yields: $\sum_{i \in T} \big(E(T) - E(T \setminus \{i\}) \big) = E(T) \implies (|T| - 1) E(T) = \sum_{i \in T} E(T \setminus \{i\})$ We proceed by induction on $m = |T|$. For $m = 1$, $T = \{i\}$ is trivially additive. Suppose $E(S) = \sum_{u \in S} E(\{u\})$ holds for all subsets of cardinality $m - 1 \ge 1$. Then for $|T| = m \ge 2$, each $T \setminus \{i\}$ has cardinality $m - 1$, so by the inductive hypothesis: $(m - 1) E(T) = \sum_{i \in T} \sum_{u \in T \setminus \{i\}} E(\{u\}) = (m - 1) \sum_{u \in T} E(\{u\})$ Since $m - 1 \ge 1$, dividing by $m - 1$ gives $E(T) = \sum_{u \in T} E(\{u\})$, completing the induction.
2. Because the combinatorial weights $w(S) \triangleq \frac{|S|!(n_w - |S|)!}{(n_w + 1)!}$ over all $S \subseteq B_w$ satisfy $\sum_{S \subseteq B_w} w(S) = \sum_{k=0}^{n_w} \binom{n_w}{k} \frac{k!(n_w - k)!}{(n_w + 1)!} = \sum_{k=0}^{n_w} \frac{1}{n_w + 1} = 1$, we can write $\Delta E_{\text{marginal}}(r_t \mid B_w) = \sum_{S \subseteq B_w} w(S) \Delta E_{\text{marginal}}(r_t \mid B_w)$. Subtracting this from $\phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) = \sum_{S \subseteq B_w} w(S) \Delta E_{\text{marginal}}(r_t \mid S)$ and noting that the $S = B_w$ term vanishes identically yields the exact identity in Part 2. Since $w(S) > 0$ for all $S \subsetneq B_w$, the strict inequalities for submodular and supermodular E follow immediately.
3. Under $E_w(S) = E_{\text{wake}, w}\mathbb{I}\{S \neq \emptyset\} + \sum_{j \in S} e_{w,j}$, for any $n_w \ge 1$ and $S \subseteq B_w$, $\Delta E_{\text{marginal}}(r_t \mid S) - \Delta E_{\text{marginal}}(r_t \mid B_w)$ equals $E_{\text{wake}, w}$ when $S = \emptyset$ (which has weight $w(\emptyset) = \frac{1}{n_w+1}$) and 0 when $S \neq \emptyset$. By Part 2, $\phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) = e_{w, r_t} + \frac{E_{\text{wake}, w}}{n_w + 1}$. Under the inequality in Part 3, $e_{w_1, r_t} < e_{w_2, r_t}$ while $e_{w_1, r_t} + \frac{E_{\text{wake}, w_1}}{n_1 + 1} > e_{w_2, r_t} + \frac{E_{\text{wake}, w_2}}{n_2 + 1}$,

proving that Shapley dispatch selects w_2 and wastes e_{w_2, r_t} - e_{w_1, r_t} > 0 Joules. \blacksquare

3.3 Repairing IW-dACI: Complete Proof of Theorem 2 and Implementation Consistency

We now resolve the IW-dACI implementation bug where updating only on audited completions (I_t = 1) while dividing by p_{\text{audit}} distorted the target risk rate by 1/p_{\text{audit}}. We prove two exact, implementation-consistent bounds: **Formulation A (All-Step Horvitz-Thompson Ring-Buffer Update)** which bounds the *unconditional population unsafe deferral rate* \frac{1}{T-h}\sum_{t=1}^{T-h} \text{err}_t \le \alpha, and **Formulation B (Audited-Step Unweighted Update)** which bounds the *conditional unsafe deferral rate among small-tier routed queries* \frac{1}{N_{\text{small}}} \sum_{t: A_t^{\text{actual}}=1} \mathbb{I}\{Y_t = 0\} \le \alpha_{\text{cond}}.

Theorem 2 (Implementation-Consistent Non-Exchangeable Safety Bounds for IW-dACI): *Let \{(x_t, Y_t)\}_{t=1}^T be any arbitrary, non-stationary, non-exchangeable sequence of requests and binary safety labels Y_t = 1 - \mathbb{I}\{m_{\text{small}}\text{ wrong}\}\mathbb{I}\{m_{\text{large}}\text{ correct}\} \in \{0, 1\}. At arrival t, let the gate eligibility be G_t = \mathbb{I}\{\hat{s}(x_t) \ge \tau_t \wedge C_{d(x_t)} \le C_{\text{max}}\} with \tau_t = \text{Quantile}(1 - \Pi_{[0,1]}(\alpha_t), \mathcal{S}_{\text{cal}}) and boundary convention \alpha_t \le 0 \implies \tau_t = +\infty \implies G_t = 0. Let the dispatcher choose worker $w_t^$, resulting in actual small-tier indicator A_t = \mathbb{I}\{m(w_t^*) = m_{\text{small}}\} \le G_t and true unsafe deferral indicator \text{err}_t \triangleq A_t \mathbb{I}\{Y_t = 0\} \in \{0, 1\}.*

*Suppose that at every step t \in \{1, \dots, T\}, the router draws I_t \sim \text{Bernoulli}(p_{\text{audit}}) (p_{\text{audit}} \in (0, 1]) independently of (x_t, Y_t, A_t, \mathcal{H}_t), executes a shadow check on m_{\text{large}} whenever A_t = 1 \wedge I_t = 1 (completing within h \ge 0 request arrivals, or conservatively substituting Y_t = 0 if still pending at step t+h), and pushes \widehat{\text{err}}_t \triangleq \frac{I_t}{p_{\text{audit}}} A_t \mathbb{I}\{Y_t = 0\} into a ring buffer of lag h (so \widehat{\text{err}}_t = 0 whenever A_t = 0 or I_t = 0). If \alpha_1 \in [0, 1] and \alpha_{t+1} is updated **at every request arrival t** via:*

$$\alpha_{t+1} = \min\!\Big(1,\; \alpha_t + \gamma \left(\alpha - \widehat{\text{err}}_{t-h} \right) \Big) \qquad \left(\text{with } \widehat{\text{err}}_s \triangleq \alpha \text{ for } s \le 0\right)$$

then for every sample path and every T > h:

1. *(State Invariant): \alpha_t \in \left[-\frac{(h+1)\gamma}{p_{\text{audit}}},\; 1 \right] for all t \ge 1.*
2. *(Deterministic Sample-Path Bound on Importance-Weighted Error):* $\frac{1}{T - h} \sum_{t=1}^{T - h} \widehat{\text{err}}_t \le \alpha + \frac{1 + (h + 1)\gamma / p_{\text{audit}}}{\gamma (T - h)}$
3. *(High-Probability Population Bound on All Routed Traffic): With probability at least 1 - \delta over \{I_t\}_{t=1}^{T-h}, the true population unsafe deferral rate across all T - h requests satisfies:* $\frac{1}{T - h} \sum_{t=1}^{T - h} \text{err}_t \le \alpha + \frac{1 + (h + 1)\gamma / p_{\text{audit}}}{\gamma (T - h)} + \frac{1}{p_{\text{audit}}} \sqrt{\frac{\log(1/\delta)}{2 (T - h)}}$
4. *(Alternative Conditional Bound on Audited Subsequence): If instead the recursion \alpha_{m+1}^{\text{cond}} = \min\!\big(1,\, \alpha_m^{\text{cond}} + \gamma(\alpha_{\text{cond}} - \mathbb{I}\{Y_{t_m - h_{\text{aud}}} = 0\})\big) is updated **only** upon audited small-tier steps m = 1, \dots, M (where A_{t_m}=1 \wedge I_{t_m}=1) using the **unweighted** error \mathbb{I}\{Y = 0\} (without dividing by p_{\text{audit}}), then*

with probability at least 1 - \delta over the i.i.d. Bernoulli(p_{\text{audit}}) coins among the N_{\text{small}} small-tier routed queries, the conditional failure rate among small-tier queries satisfies \frac{1}{N_{\text{small}}}\sum_{t: A_t=1} \mathbb{I}\{Y_t = 0\} \le \alpha_{\text{cond}} + O\!\left(\frac{1}{\gamma p_{\text{audit}} N_{\text{small}}} + \sqrt{\frac{\log(1/\delta)}{p_{\text{audit}} N_{\text{small}}}}\right).

Complete Proof:

- **Proof of Parts 1–3:** Because A_t \le G_t, whenever \alpha_s \le 0 we have G_s = 0 \implies A_s = 0 \implies \text{err}_s = 0 \implies \widehat{\text{err}}_s = 0. If \alpha_{t_0} > 0 and \alpha_{t_0+1} \le 0, at most h delayed feedback values from steps \{t_0 - h + 1, \dots, t_0\} can be positive (each at most 1/p_{\text{audit}}), after which every subsequent feedback \widehat{\text{err}}_s arriving while \alpha \le 0 is identically 0 and increases \alpha by +\gamma \alpha > 0. Thus \alpha_t \ge -\frac{(h+1)\gamma}{p_{\text{audit}}} and \alpha_t \le 1 for all t \ge 1. Writing \alpha_{t+1} - \alpha_t = \gamma(\alpha - \widehat{\text{err}}_{t-h}) - \xi_t with upper-clipping slack \xi_t = \max(0, \alpha_t + \gamma(\alpha - \widehat{\text{err}}_{t-h}) - 1) \ge 0 and summing over t = h+1, \dots, T yields \alpha_{T+1} - \alpha_{h+1} \le \gamma(T-h)\alpha - \gamma \sum_{s=1}^{T-h} \widehat{\text{err}}_s, which rearranges to Part 2 using \alpha_{h+1} - \alpha_{T+1} \le 1 + \frac{(h+1)\gamma}{p_{\text{audit}}}. Finally, conditioned on \mathcal{F}_s (which contains x_s, Y_s, A_s, and thus \text{err}_s), \mathbb{E}[\widehat{\text{err}}_s \mid \mathcal{F}_s] = \frac{\mathbb{E}[I_s \mid \mathcal{F}_s]}{p_{\text{audit}}} \text{err}_s = \text{err}_s, and D_s = \text{err}_s - \widehat{\text{err}}_s has conditional range at most 1/p_{\text{audit}} (since D_s = 0 when \text{err}_s = 0 and D_s \in [1 - 1/p_{\text{audit}}, 1] when \text{err}_s = 1). Applying the one-sided Azuma-Hoeffding inequality to \sum_{s=1}^{T-h} D_s yields Part 3. Note also that if an audit at step s exceeds lag h and substitutes \widehat{\text{err}}_s^{\text{conserv}} = 1/p_{\text{audit}} \ge \widehat{\text{err}}_s, then \sum \text{err}_s \le \sum \widehat{\text{err}}_s + \sum D_s \le \sum \widehat{\text{err}}_s^{\text{conserv}} + \sum D_s, so the one-sided upper safety bound remains 100% valid!
- **Proof of Part 4 (and Counterexample to Turn 2's Pseudocode):** Among the subsequence of steps \{t_k\}_{k=1}^{N_{\text{small}}} where A_{t_k} = 1, I_{t_k} \sim \text{Bernoulli}(p_{\text{audit}}) selects M \approx p_{\text{audit}} N_{\text{small}} audited steps. Updating only on those M audited steps with the unweighted error \text{err}_m^{\text{unw}} = \mathbb{I}\{Y_{t_m} = 0\} \in \{0, 1\} telescopes over M to give \frac{1}{M - h_{\text{aud}}}\sum_{m=1}^{M - h_{\text{aud}}} \mathbb{I}\{Y_{t_m} = 0\} \le \alpha_{\text{cond}} + \frac{1 + (h_{\text{aud}}+1)\gamma}{\gamma(M - h_{\text{aud}})}, which concentrates around \frac{1}{N_{\text{small}}}\sum_{k=1}^{N_{\text{small}}} \mathbb{I}\{Y_{t_k}=0\} by Azuma-Hoeffding since \mathbb{E}[I_{t_k} \mathbb{I}\{Y_{t_k}=0\} \mid \mathcal{F}_{t_k}] = p_{\text{audit}} \mathbb{I}\{Y_{t_k}=0\}. **Counterexample if divided by p_{\text{audit}} while updating only on I_{t_m}=1:** If \text{err}_m^{\text{unw}} is divided by p_{\text{audit}} inside an audit-only update loop, the telescope forces \frac{1}{M}\sum_{m=1}^M \frac{\mathbb{I}\{Y_{t_m}=0\}}{p_{\text{audit}}} \le \alpha_{\text{cond}} \implies \frac{1}{M}\sum_{m=1}^M \mathbb{I}\{Y_{t_m}=0\} \le p_{\text{audit}} \alpha_{\text{cond}} (e.g., 0.05 \times 0.05 = 0.0025), driving \alpha_t \to 0 and shutting down small-tier routing even when m_{\text{small}} is 99\% accurate! \blacksquare

3.4 Routing-Objective Assumptions and Explicit Proof Gap

The PIGOU-J dispatch objective at arrival t:
w^*(r_t) = \arg\min_{w \in \mathcal{W}_{\text{feas}}(r_t)} \Big\{ \Delta \hat{E}_{\text{self}}(r_t \mid s_w) + \beta_E \Delta \hat{E}_{\text{ext}}(r_t \to B_w \mid s_w) + \lambda_L(t) \big(\Delta \hat{L}_{\text{self}}(r_t \mid s_w) + \beta_L \Delta \hat{L}_{\text{ext}}(r_t \to B_w \mid s_w) \big) \Big\}
rests on four explicit mathematical assumptions:

1. **Exactness at \beta_E = 1:** By Section 3.1, \Delta \hat{E}_{\text{self}} + \beta_E \Delta \hat{E}_{\text{ext}} equals the exact open-loop counterfactual marginal worker energy \Delta \hat{E}_{\text{marginal}}(r_t \mid s_w) *if and only if* \beta_E = 1. Any \beta_E \neq 1 (specifically \beta_E = 0) is strictly an ablation configuration to measure the penalty of ignoring step-elongation energy externalities.
2. **Additive Virtual Queue Evolution for \lambda_L(t):** To connect \lambda_L(t) to Lyapunov drift-plus-penalty, the latency deficit state Z_L(t) \ge 0 [\text{s}] must evolve via an additive virtual queue recursion updated upon request completions j (or per-epoch predicted arrivals): Z_L(t^+) = \max\!\Big(0,\; Z_L(t) + \big(\text{TTFT}_j - L^{\text{TTFT}}\big) + \omega_D \big(\text{TPOT}_j - L^{\text{TPOT}}\big)\Big), \qquad \lambda_L(t) \triangleq \frac{1}{V_J} Z_L(t) where V_J > 0 [\text{s}^2/\text{J}] is the Lyapunov penalty parameter (so \lambda_L(t) = Z_L(t)/V_J has exact units of [\text{J/s} = \text{W}]).
3. **Conditional-Survival Stationarity Assumption:** The remaining decode step estimator \tilde{d}_{w,\text{rem}} = \frac{1}{n_w}\sum_{j \in B_w} \mathbb{E}[d_j - d_{j,\text{elap}} \mid d_j > d_{j,\text{elap}}, \hat{d}_j] assumes the conditional hazard distribution of decode lengths changes slowly relative to the sliding Kaplan-Meier window (W_{\text{surv}} = 500 completions).
4. **Explicit Proof Gap (Coupled Rollout Trajectory Drift \Xi_T):** Because each routing action w_t^* modifies the batch state s_{w_t^*}(t') and KV-cache residency \mathcal{C}_{w_t^*}(t') encountered by future arrivals t' > t, minimizing the one-step counterfactual marginal cost \Delta \hat{E}_{\text{marginal}}(r_t \mid s_w(t)) is a **myopic policy** with respect to the coupled multi-step Markov Decision Process (MDP). We explicitly do *not* claim a proof that myopic PIGOU-J is globally optimal over infinite-horizon coupled sample paths; instead, we bound single-step decision regret in Exp 2 and measure multi-step rollout performance empirically in Exp 3 and Exp 4.

3.5 Independently Recalculated Statistical Power and Sample-Size Table

Using the exact non-central t-distribution (scipy.stats.nct) and the exact iterative t-quantile formula for a two-sided paired test at family-wise error rate \alpha_{\text{FWER}} = 0.05 across K_{\text{comp}} = 4 primary baselines (\alpha_1 = 0.05 / 4 = 0.0125, two-sided critical level \alpha_1/2 = 0.00625, z_{1-\alpha_1/2} = 2.4977), target statistical power 1 - \beta = 0.90 (z_{1-\beta} = 1.2816), and minimum practically meaningful effect \delta_{\text{min}} = 4.0\% (0.04), the table below reports the exact required number of independent paired-seed runs N across empirical paired standard deviations \hat{\sigma}_{\text{diff}} \in [1.5\%, 5.0\%], including the mandatory Wilcoxon signed-rank Asymptotic Relative Efficiency (\text{ARE}) adjustments (N_{\text{Wilc, norm}} = \lceil N_t / 0.955 \rceil for normal errors and N_{\text{Wilc, worst}} = \lceil N_t / 0.864 \rceil for worst-case symmetric continuous distributions):

Paired Run-to-Run SD (sigma_diff)	Exact Paired t-Test Sample Size (N_t)	Exact Non-Central t Power at N_t	Wilcoxon Required Seeds (Normal ARE = 0.955)	Wilcoxon Required Seeds (Worst-Case Symmetric ARE = 0.864)	Operational Decision in Exp 0
1.5%	6	0.966	7	7	Proceed with N = 8 paired seeds
2.0%	7	0.910	8	9	Proceed with N = 9 paired seeds
2.5%	9	0.907	10	11	Proceed with N = 11 paired seeds
3.0%	12	0.925	13	14	Proceed with N = 14 paired seeds
3.5%	15	0.922	16	18	Proceed with N = 18 paired seeds
4.0%	18	0.911	19	21	High variance; lock fans & increase window to 20 min
5.0%	26	0.907	28	31	Trigger Kill Criterion K1 (halt until thermal/noise fixed)

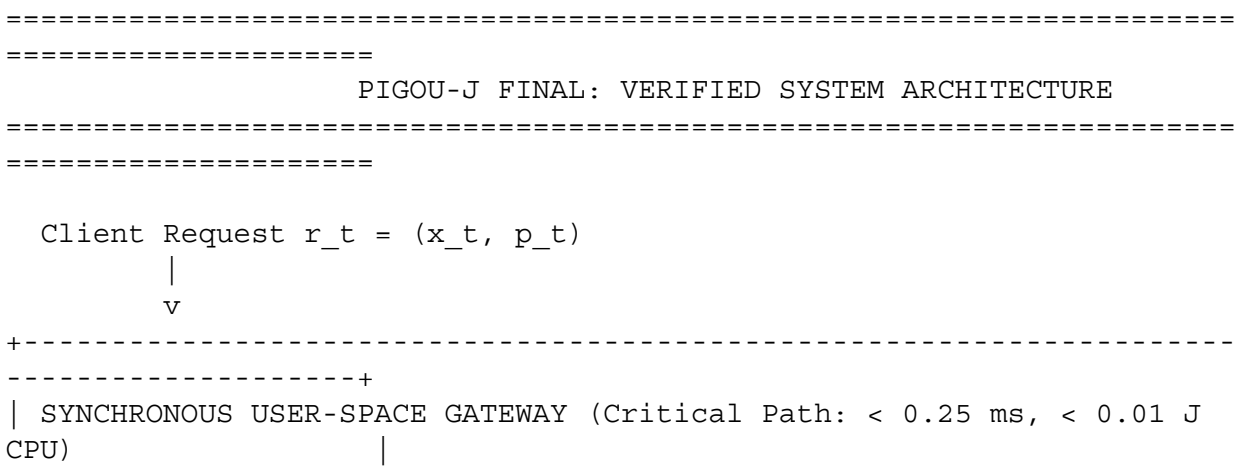
4. Final Minimized Architecture (PIGOU-J Final, Requirement 4)

4.1 Component Retention, Simplification, and Removal Decisions

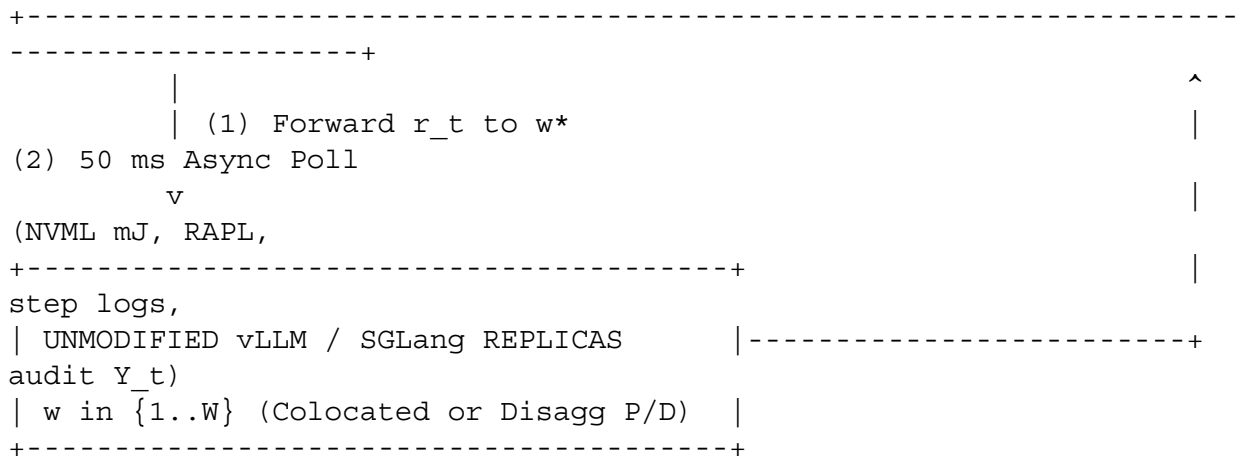
Component	Status in PIGOU-J Final	Justification
1. Non-Overlapping Unclipped Marginal-Joule & Externality Scorer (PIGOU-J Final)	**CORE CONTRIBUTION**	Prices exact counterfactual \Delta \hat{E}_{\text{self}} + \beta_E \Delta \hat{E}_{\text{ext}} and \lambda_L(t)(\Delta \hat{L}_{\text{self}} + \beta_L \Delta \hat{L}_{\text{ext}}) with zero clipping distortion and zero prefill-decode step overlap double-counting.
2. Ring-Buffer Consistent	**CORE CONTRIBUTION**	Updates \alpha_{t+1} = \min(1,

Component	Status in PIGOU-J Final	Justification
IW-dACI Gate	**(2-Tier Mode)**	\alpha_t + \gamma(\alpha - \widehat{\text{err}}_{t-h})) on every request t via a length-h ring buffer, matching Theorem 2 line-for-line.
3. Sliding Kaplan-Meier Survival Estimator (\tilde{d}_{w,\text{rem}})	**REQUIRED INFRASTRUCTURE**	Computes \mathbb{E}[d - d_{\text{elap}} \mid d > d_{\text{elap}}, \hat{d}_j] over the last W_{\text{surv}} = 500 completions so heavy-tailed outputs update remaining batch horizons dynamically.
4. In-Memory Prefix-Hash Mirror & In-Flight Counter	**REQUIRED INFRASTRUCTURE**	Computes 16-token block hashes to obtain resident prefix tokens p_{t,w}^{\text{hit}} and tracks unacknowledged in-flight tokens without external etcd daemons or k_b LPT claims.
5. Fast Input Length & Safety Head	**REQUIRED INFRASTRUCTURE**	Predicts (\hat{d}_t, \hat{s}(x_t)) in <0.20\text{ ms} on CPU (LightGBM / logistic head on token/embedding features).
6. Node-Level Phase DVFS (PowerSlider / Tri-serve)	**DEFERRED TO ABLATION A5**	Tested strictly as an orthogonal node-level substrate beneath all routers in Ablation A5; never claimed as part of our invention.
7. Ex-Post Shapley Dispatch (JCalib), WAN Carbon Shifting, & Cascades	**REMOVED**	JCalib is retained strictly as Baseline B4; WAN shifting and sequential cascades are permanently excluded.

4.2 Text Architecture Diagram and Request Lifecycle



```
|
|
|  [Stage 1: Token, Prefix & Survival State Inspector]
|
|     |-- 16-token block hash lookup in local Radix mirror ->
p_{t,w}^hit                  |
|     |-- Effective uncached prefill tokens: p_{t,w}^eff = p_t -
p_{t,w}^hit              |
|     |-- Fast predictor -> decode length d_hat_t, safety score
s_hat(x_t), domain C_d    |
|
|
|  [Stage 2: Ring-Buffer Consistent IW-dACI Gate (Active in 2-Tier
Mode Only)]             |
|     |-- Pops slot (t - h) from ring buffer -> updates alpha_t =
min(1, alpha_{t-1} +    |
|         gamma * (alpha_target - err_hat_{t-h})) on EVERY arrival t
(Theorem 2)          |
|     |-- If alpha_t <= 0 or C_d > C_max or s_hat(x_t) < tau_t: W_feas
= {m_large}        |
|     |-- Else: W_feas = {m_small, m_large}
|
|
|
|  [Stage 3: Non-Overlapping Unclipped PIGOU-J Dispatcher]
|
|     For each w in W_feas(r_t) with state s_w = (n_w, Q_w, M_w,
d_tilde_{w,rem}):        |
|     |-- Stage I/II deducts K_wait and K_pre -> post-prefill
d_rem^(2)                    |
|     |-- Computes N_ov = I{n_w>0}*min(d_hat_t, d_rem^(2)),  N_solo =
d_hat_t - N_ov      |
|     |-- Unclipped Private Marginal Joules:   dE_self(r_t | s_w)
[J]               |
|     |-- Unclipped Incumbent Energy External: dE_ext(r_t -> B_w |
s_w) [J]               |
|     |-- Additive Virtual Queue Price:        lambda_L(t) =
Z_L(t)/V_J [W = J/s]         |
|     |-- Routes r_t to w* = argmin_{w in W_feas} { dE_self +
beta_E*dE_ext               |
|                                                    +
lambda_L(t)*(dL_self+beta_L*dL_ext)}|
|     |-- Pushes slot t into IW-dACI ring buffer; if m(w*) == m_small,
draws i.i.d.       |
|         I_t ~ Bernoulli(p_audit): if I_t == 0 sets slot t = 0.0
immediately;            |
|         if I_t == 1 dispatches async shadow check to fill slot t
with err_t / p_audit   |
```



4.3 Definitive Verified Algorithm Pseudocode (PIGOU-J Final)

```
# ALGORITHM: PIGOU-J Final (Unclipped Non-Overlapping Marginal Joules
+ All-Step Ring-Buffer IW-dACI)
# Complexity: O(|W|) arithmetic ops per arrival (< 0.05 ms on CPU);
zero clipping distortion.

class IWDACIRingBuffer:
    def __init__(self, lag_h: int, alpha_target: float, gamma: float,
p_audit: float, cal_scores):
        self.h = max(1, lag_h)
        self.alpha_target = alpha_target
        self.gamma = gamma
        self.p_audit = p_audit
        self.alpha_t = alpha_target
        self.cal_scores = cal_scores
        self.tau_t = empirical_quantile(self.cal_scores, 1.0 -
self.alpha_t)
        # Ring buffer holds resolved err_hat floats; None indicates
pending in-flight audit
        self.buf = [alpha_target for _ in range(self.h)]
        self.step_idx = 0

    def step_and_get_threshold(self) -> tuple[float, float]:
        slot = self.step_idx % self.h
        err_hat_delayed = self.buf[slot]
        if err_hat_delayed is None:
            # Conservative fail-safe if audit did not finish within h
arrivals (Theorem 2 proof)
            err_hat_delayed = 1.0 / self.p_audit
        # Exact one-sided upper-clipped recursion executed on EVERY
arrival t
        self.alpha_t = min(1.0, self.alpha_t + self.gamma *
```

```
(self.alpha_target - err_hat_delayed))
        if self.alpha_t <= 0.0:
            self.tau_t = float("inf")
        else:
            self.tau_t = empirical_quantile(self.cal_scores, 1.0 -
self.alpha_t)
        return self.alpha_t, self.tau_t

    def register_arrival_outcome(self, routed_to_small: bool) ->
tuple[int, bool]:
        t_cur = self.step_idx
        slot = t_cur % self.h
        self.step_idx += 1
        if not routed_to_small:
            # Routed to LARGE: true unsafe deferral error is
deterministically 0.0
            self.buf[slot] = 0.0
            return t_cur, False
        # Routed to SMALL: draw independent Bernoulli(p_audit) coin
        if random_uniform_0_1() < self.p_audit:
            self.buf[slot] = None  # Pending async shadow audit
resolution
            return t_cur, True
        else:
            # Unaudited step: Horvitz-Thompson estimator is (0 /
p_audit) * err_t = 0.0
            self.buf[slot] = 0.0
            return t_cur, False

    def resolve_audit(self, arrival_step_id: int, small_correct: bool,
large_correct: bool):
        if self.step_idx - arrival_step_id <= self.h:
            err_realized = 1.0 if (not small_correct and
large_correct) else 0.0
            self.buf[arrival_step_id % self.h] = err_realized /
self.p_audit


def pigou_j_final_dispatch(req, workers, prof, state, iwdaci:
IWDACIRingBuffer):
    p_t = len(req.token_ids)
    d_hat, s_hat, domain_C = state.predictor.predict(req.token_ids)

    # 1. Advance IW-dACI by 1 arrival step and determine
Conformal-feasible workers
    if state.two_tier_mode:
        alpha_cur, tau_cur = iwdaci.step_and_get_threshold()
        if alpha_cur <= 0.0 or domain_C > state.C_max or s_hat <
```

```
tau_cur:
            feas = [w for w in workers if w.tier == "LARGE" and
w.can_admit(p_t + d_hat)]
        else:
            feas = [w for w in workers if w.can_admit(p_t + d_hat)]
    else:
        feas = [w for w in workers if w.can_admit(p_t + d_hat)]

    if not feas:
        best_w = fallback_least_inflight_tokens(workers)
        t_id, audit_flag =
(iwdaci.register_arrival_outcome(best_w.tier == "SMALL")
                            if state.two_tier_mode else (0, False))
        return best_w, t_id, audit_flag

    best_w = None
    best_joule_score = float("inf")
    lambda_L_watts = state.Z_L_seconds / state.V_J

    # 2. Evaluate Non-Overlapping Unclipped Counterfactual Marginal
Joules & Delay Externality
    for w in feas:
        p_hit = state.prefix_mirror.match(w.id, req.token_ids) if
w.free_kv_ratio > 0.05 else 0
        p_eff = max(0, p_t - p_hit)
        n_w = w.active_and_inflight_reqs
        Q_w = w.queued_prefill_tokens
        M_w = w.resident_kv_tokens

        # Stage I: Queue wait before r_t prefill starts
        t_wait = prof.tau_pre(w.model, Q_w, M_w) if Q_w > 0 else 0.0
        K_wait = prof.decode_steps_during_wait(w.model, Q_w, n_w, M_w)
if n_w > 0 else 0.0
        d_rem_0 = w.survival_expected_rem_steps() if n_w > 0 else 0.0
        d_rem_1 = max(0.0, d_rem_0 - K_wait)

        # Stage II: Prefill execution of p_eff uncached tokens
(Colocated Chunked vs Dedicated)
        if w.chunk_mix and n_w > 0 and d_rem_1 > 0.0 and p_eff > 0:
            K_pre = min(d_rem_1, math.ceil(p_eff / w.chunk_size))
            p_mixed = min(p_eff, int(K_pre * w.chunk_size))
            p_rem_ded = max(0, p_eff - p_mixed)

            t_dec_base_pre = K_pre * prof.tau_dec(w.model, n_w, M_w)
            P_dec_base_pre = prof.P_dec(w.model, n_w, M_w)
            t_mix_pre = prof.tau_mix(w.model, p_mixed, n_w, M_w)
            P_mix_pre = prof.P_mix(w.model, p_mixed, n_w, M_w)
```

```
            t_ded = prof.tau_pre(w.model, p_rem_ded, M_w) if p_rem_ded
> 0 else 0.0
            P_ded = prof.P_pre(w.model, p_rem_ded, M_w) if p_rem_ded >
0 else 0.0

            # Exact unclipped bilinear split for Stage II
            dE_pre_self = (P_mix_pre - P_dec_base_pre) *
t_dec_base_pre + P_ded * t_ded
            dE_pre_ext = P_mix_pre * (t_mix_pre - t_dec_base_pre)
            t_pre_total = t_mix_pre + t_ded
            dt_pre_delay = (t_mix_pre - t_dec_base_pre) + t_ded
            d_rem_2 = max(0.0, d_rem_1 - K_pre)
        else:
            t_pre_total = prof.tau_pre(w.model, p_eff, M_w)
            P_pre_ded = prof.P_pre(w.model, p_eff, M_w)
            dE_pre_self = P_pre_ded * t_pre_total
            dE_pre_ext = 0.0
            dt_pre_delay = t_pre_total if (not w.is_disagg and n_w >
0) else 0.0
            d_rem_2 = d_rem_1

        # Stage III & IV: Post-prefill Co-Batched Overlap (N_ov) and
Solo Tail (N_solo)
        N_ov = min(float(d_hat), d_rem_2) if n_w > 0 else 0.0
        N_solo = max(0.0, float(d_hat) - N_ov)

        M_w_plus = M_w + p_t + 0.5 * N_ov
        M_solo = p_t + N_ov + 0.5 * N_solo

        t_dec_0 = prof.tau_dec(w.model, n_w, M_w) if n_w > 0 else 0.0
        P_dec_0 = prof.P_dec(w.model, n_w, M_w) if n_w > 0 else 0.0
        t_dec_plus = prof.tau_dec(w.model, n_w + 1, M_w_plus)
        P_dec_plus = prof.P_dec(w.model, n_w + 1, M_w_plus)

        t_dec_solo = prof.tau_dec(w.model, 1, M_solo)
        P_dec_solo = prof.P_dec(w.model, 1, M_solo)

        # Unclipped Signed Private Marginal Energy [Joules]
        dE_self = (dE_pre_self
                   + N_ov * (P_dec_plus - P_dec_0) * t_dec_0
                   + N_solo * P_dec_solo * t_dec_solo)

        # Unclipped Signed Pigouvian Energy Externality [Joules]
        dE_ext = dE_pre_ext + N_ov * P_dec_plus * (t_dec_plus -
t_dec_0)

        # Private Latency & Incumbent Delay Externality [Seconds]
        dL_self = t_wait + t_pre_total + N_ov * t_dec_plus + N_solo *
```

```
t_dec_solo
        dL_ext = n_w * (dt_pre_delay + N_ov * (t_dec_plus - t_dec_0))

        # Total Objective in Joules [J]
        score_J = (dE_self + state.beta_E * dE_ext) + lambda_L_watts *
(dL_self + state.beta_L * dL_ext)
        if score_J < best_joule_score:
            best_joule_score = score_J
            best_w = w

    best_w.record_inflight(p_eff, d_hat)
    if state.two_tier_mode:
        t_id, audit_flag = iwdaci.register_arrival_outcome(best_w.tier
== "SMALL")
        if audit_flag:
            enqueue_shadow_audit_on_large(req, t_id, best_w.id)
    else:
        t_id, audit_flag = 0, False

    return best_w, t_id, audit_flag
```

5. Executable Staged Validation Protocol, Corrected Energy Reconciliation, and MVP Kill Criteria (Requirement 3)

5.1 Repaired Time-Slice Energy Reconciliation vs. Counterfactual Prefix Replay

To eliminate the mathematical contradiction of summing open-loop dispatch predictions over a continuous stream, we establish two strictly distinct verification protocols:

1. **Protocol A — Continuous-Stream Time-Slice Energy Reconciliation (Exp 0, Exp 3, Exp 4, Exp 6):** Over any steady-state evaluation window $[T_{\text{start}}, T_{\text{end}}]$ of length $\Delta T = T_{\text{end}} - T_{\text{start}}$, let $\{[\tau_{k-1}, \tau_k]\}_{k=1}^K$ be the non-overlapping scheduler steps (or 50 ms telemetry frames) executed on worker $w \in \mathcal{W}$, with duration $\Delta \tau_k = \tau_k - \tau_{k-1}$. Total hardware energy measured at the node boundary is: $E_{\text{measured}}([T_{\text{start}}, T_{\text{end}}]) = \sum_{g=1}^G \kappa_g \Big(E_{\text{NVML}, g}(T_{\text{end}}) - E_{\text{NVML}, g}(T_{\text{start}}) \Big) + \Big(E_{\text{RAPL}}(T_{\text{end}}) - E_{\text{RAPL}}(T_{\text{start}}) \Big)$ where $\kappa_g = E_{\text{PDU}, g} / E_{\text{NVML}, g}$ is the external rack PDU / IPMI calibration multiplier. The reconstructed model energy over the exact same time window $[T_{\text{start}}, T_{\text{end}}]$ is integrated over the **realized non-overlapping time slices**: $\widehat{E}_{\text{slice-sum}}([T_{\text{start}}, T_{\text{end}}]) = \left(\sum_{g=1}^G P_{\text{idle}, g} + P_{\text{CPU, idle}} \right) \Delta T + E_{\text{RAPL, active}} + \sum_{w=1}^W \sum_{k=1}^K P_{\text{model}, w}\!\left(n_{w,k}, p_{w,k}^{\text{eff}}, M_{w,k}\right) \Delta \tau_k$ Because $\bigcup_{k=1}^K [\tau_{k-1}, \tau_k] = [T_{\text{start}}, T_{\text{end}}]$ is a partition of time, requests straddling T_{start} or T_{end} have *only* their in-window slices $[\tau_{k-1}, \tau_k] \subseteq$

[T_{\text{start}}, T_{\text{end}}] included in both E_{\text{measured}} and \widehat{E}_{\text{slice-sum}}, eliminating boundary-straddle estimation errors completely. The reconciliation residual is: \epsilon_{\text{rec}} \triangleq \frac{\widehat{E}_{\text{slice-sum}}([T_{\text{start}}, T_{\text{end}}]) - E_{\text{measured}}([T_{\text{start}}, T_{\text{end}}])}{E_{\text{measured}}([T_{\text{start}}, T_{\text{end}}])} which must satisfy |\epsilon_{\text{rec}}| \le 0.02 (2.0\%).

2. **Protocol B — Fixed-Horizon Counterfactual Busy-Period Prefix Replay (Exp 2):** To validate the ex-ante counterfactual estimator \Delta \hat{E}_{\text{marginal}}(r_m \mid s_w) directly, we sample M = 100 deterministic 2-worker states (B_{w_1}, B_{w_2}) and a newcomer r_m. For each scenario m \in \{1, \dots, 100\}, we fix a common horizon H_m = 6.0\text{ s} (guaranteed to exceed maximum drain time) and execute three isolated runs from a clean idle state (R = 5 hardware repeats each at temperature = 0):
    - **Run 0 (Baseline):** Execute (B_{w_1}, B_{w_2}) with no newcomer over [0, H_m], recording total active energy E_{\text{act}}^{(0)}.
    - **Run 1 (Place on w_1):** Execute (B_{w_1} \cup \{r_m\}, B_{w_2}) over [0, H_m], recording E_{\text{act}}^{(1)} and empirical marginal increment \widehat{\Delta E}_{\text{replay}}(r_m \mid s_{w_1}) = \bar{E}_{\text{act}}^{(1)} - \bar{E}_{\text{act}}^{(0)}.
    - **Run 2 (Place on w_2):** Execute (B_{w_1}, B_{w_2} \cup \{r_m\}) over [0, H_m], recording E_{\text{act}}^{(2)} and empirical marginal increment \widehat{\Delta E}_{\text{replay}}(r_m \mid s_{w_2}) = \bar{E}_{\text{act}}^{(2)} - \bar{E}_{\text{act}}^{(0)}. Because no subsequent requests arrive after r_m during [0, H_m], \widehat{\Delta E}_{\text{replay}}(r_m \mid s_{w_r}) is an unbiased physical measurement of \Delta E_{\text{marginal}}(r_m \mid s_{w_r}) with standard error \text{SE}_m = \sqrt{\text{Var}(E_{\text{act}}^{(r)})/R + \text{Var}(E_{\text{act}}^{(0)})/R}.

5.2 Corrected Experiment-Duration and Practical Cost Budget

Every single trace-replay run requires 3\text{ min} warm-up + 15\text{ min} steady-state measurement + 2\text{ min} drain + 5\text{ min} thermal cooldown reset (to \le 38^\circ\text{C} idle temperature and baseline idle watts) = **25\text{ minutes per run}**. The corrected wall-clock budget across all validation stages is:

Validation Stage	Exact Run / Replay Breakdown	Per-Unit Duration	Total Wall-Clock Time	GPU-Hours (on 4-GPU Node)
Stage 0: AA Variance & Calibration Pilot (Exp 0)	N_0 = 6 paired AA runs of SQF (12 single runs)	25\text{ min/run}	**5.0 hours**	20.0 GPU-h
Stage 1A: Factored Step Surface Profiler (Exp 1)	57 grid points (15 prefill, 27 decode, 15 mixed) \times 5 repeats	25\text{ s/point}	**1.2 hours**	4.8 GPU-h
Stage 1B: Counterfactual Prefix Replay	M = 100 scenarios \times 3 arms \times R = 5	8\text{ s/replay}	**3.3 hours**	6.7 GPU-h (2 GPUs)

Validation Stage	Exact Run / Replay Breakdown	Per-Unit Duration	Total Wall-Clock Time	GPU-Hours (on 4-GPU Node)
(Exp 2)	repeats = 1,500 replays			
Stage 2A: Fast Screening MVP (Exp 3-MVP)	3 core policies (PIGOU-J, MARCS-SC, JCalib) \times 2 loads \times 8 seeds = 48 runs	25\text{ min/run}	**20.0 hours**	80.0 GPU-h
Stage 2B: Full Homogeneous Evaluation (Exp 3-Full)	5 policies \times 2 loads \times 11 seeds (at \sigma_{\text{diff}} =2.5\%) = 110 runs (minus 48 MVP runs = 62 additional runs)	25\text{ min/run}	**25.8 hours** (45.8 h total)	103.3 GPU-h (183.3 GPU-h total)
Stage 3A: CPU-Only IW-dACI Drift Replay (Exp 5)	4 conformal policies \times 20 seeds \times 2,000 pre-scored queries	3\text{ s/seed}	**0.1 hours**	0 GPU-h (CPU only)
Stage 3B: Two-Tier Heterogeneous Evaluation (Exp 4)	5 policies \times 10 seeds = 50 runs on 4–8 GPUs	25\text{ min/run}	**20.8 hours**	83.3–166.7 GPU-h

Note that Stages 0, 1A, 1B, and 2A (the complete Minimum Viable Pilot) require exactly 5.0 + 1.2 + 3.3 + 20.0 = \mathbf{29.5\text{ hours}} of wall-clock execution—fitting comfortably inside a 48-hour pilot window, whereas the full 110-run Exp 3 + 50-run Exp 4 suite requires an additional 46.7\text{ hours} (\sim 2\text{ days}) once the MVP passes.

5.3 Minimum Viable Pilot (MVP) Protocol and Quantitatively Justified Kill Criteria

Execute Stages 0, 1A, 1B, and 2A in sequence on a 4-GPU node (e.g., 4× NVIDIA A100-80GB or 4× H100-80GB) running vLLM v0.6.x/v1 or SGLang v0.4+ serving Qwen2.5-7B-Instruct:

- **Kill Criterion K1 (Stage 0 Telemetry & Repeatability Gate — Hour 5):** If the Stage 0 AA pilot measures a paired run-to-run standard deviation \hat{\sigma}_{\text{diff}} > 4.0\% or a time-slice energy reconciliation residual |\epsilon_{\text{rec}}| > 2.5\% after locking GPU cooling fans and enforcing the 38^\circ\text{C} cooldown gate, **halt execution immediately**. Do not run routing comparisons until thermal/power drift is stabilized.
- **Kill Criterion K2 (Stage 1B Counterfactual Regret Gate — Hour 9.5):** On the M = 100 deterministic 2-worker busy-period prefix replays (Exp 2), define the single-step physical decision regret of policy \pi on scenario m as R_m(\pi) \triangleq \widehat{\Delta E}_{\text{replay}}(r_m \mid s_{w_\pi^*(m)}) - \min_{w \in \{w_1, w_2\}} \widehat{\Delta E}_{\text{replay}}(r_m \mid s_w). Let \bar{\sigma}_{\text{rep}} = \sqrt{\frac{1}{100}\sum_{m=1}^{100} \text{SE}_m^2} be the empirical replay noise floor. If

PIGOU-J Final fails to achieve a lower mean decision regret \bar{R}(\text{PIGOU-J}) than both Token-Proportional and JCalib-Shapley by at least 2\bar{\sigma}_{\text{rep}} ($p < 0.01$, two-sided paired Wilcoxon test across M=100 scenarios), **abandon the PIGOU-J marginal energy estimator immediately**.

- **Kill Criterion K3 (Stage 2A Homogeneous End-to-End Gate — Hour 29.5):** Across the 8 paired seeds of Stage 2A comparing PIGOU-J Final, MARCS-SC, and JCalib-Shapley at \rho = 0.30 (diurnal light load testing n_w = 0 \to 1 wake-up consolidation) and \rho = 0.70 (shared-prefix load testing KV-hit prefill Watts vs. queue delay), if PIGOU-J Final fails to reduce **Total Node Joules per 1,000 Goodput Tokens** by at least \delta_{\text{min}} = 4.0\% over MARCS-SC in *at least one* of the two load regimes (or if P99 TTFT / P99 TPOT degrades by > 5.0\%), **drop all homogeneous routing claims** and either restrict the contribution strictly to heterogeneous two-tier routing (Stage 3B, requiring \ge 7.0\% net Joule reduction including shadow-audit energy) or abandon the project.

6. Full Evaluation Plan, Baseline Matrix, Ablations, and Evidence Map (Requirements 3 & 4)

6.1 Normalized Baseline Matrix

All baselines execute on the exact same physical GPUs, same SGLang/vLLM version, same seeded BurstGPT + LMSYS-Chat-1M / LiveBench / LiveCodeBench arrival trace, and same TTFT/TPOT SLO targets:

Baseline ID	Policy Name	Exact Mathematical Dispatch Rule	Hypothesis Falsified if Baseline Wins
B1	SQF + FIFO	\arg\min_w n_w(t)	Falsifies need for token- or energy-aware state modeling.
B2	JSRT (Remaining Tokens)	\arg\min_w \big(Q_w(t) + p_{t,w}^{\text{eff}} + n_w(t)\tilde{d}_{w,\text{rem}} + \hat{d}_t\big)	Falsifies need for calibrated step duration and power surfaces.
B3	Token-Proportional Energy	\arg\min_w \big\{ e_{m(w)}(p_{t,w}^{\text{eff}} + \hat{d}_t) + \lambda_L(t)\Delta \hat{L}_{\text{self}} \big\}	Falsifies claim that batch-state non-linearity matters beyond token count.
B4	JCalib-Shapley Dispatch	\arg\min_w \big\{ \hat{\phi}_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) + \lambda_L(t)\Delta \hat{L}_{\text{self}} \big\}	Directly tests **Theorem 1** (Ex-Post Shapley vs. Ex-Ante Marginal Cost).
B5	MARCS-SC (Latency Externality)	\arg\min_w \big\{ \Delta \hat{L}_{\text{self}}(r_t \mid s_w) + \beta_L	Tests whether pricing physical Joules (\Delta \hat{E}_{\text{self}} +

Baseline ID	Policy Name	Exact Mathematical Dispatch Rule	Hypothesis Falsified if Baseline Wins
		\Delta \hat{L}_{\text{ext}}(r_t \to B_w \mid s_w) \big\}	\Delta \hat{E}_{\text{ext}}) improves upon latency-only externality routing.
B6	Static Conformal + Token Cost	BGE logistic gate with static split-conformal t^* (2026.acl-srw.70) + B3	Tests whether online IW-dACI and marginal Joule pricing are needed in 2-tier routing.
B7	Audit-Only Naive dACI	Updates \alpha_t only on I_t=1 while dividing by p_{\text{audit}} (or trough-only auditing)	Demonstrates the empirical failure of the un-repaired Turn 2 audit rule (Theorem 2 Part 4).
B8	Myopic Hindsight Oracle	Uses realized ground-truth d_t, realized Y_t, and exact step energy	Quantifies the upper bound and single-step estimation gap \sum_t \Delta_{\text{est}}(t).

6.2 Orthogonal Ablation Suite

Ablation ID	Exact Modification to PIGOU-J Final	Physical Mechanism Isolated
A1: Selfish Marginal Energy (beta_E = 0, beta_L = 0)	Sets \beta_E = 0 and \beta_L = 0 (optimizes \Delta \hat{E}_{\text{self}} + \lambda_L(t)\Delta \hat{L}_{\text{self}} only).	Quantifies the exact Joule and P99 TPOT penalty of ignoring Pigouvian externalities on co-batched incumbents.
A2: Clipping Distortion Check	Re-introduces max(0.0, P_dec_plus - P_dec_0) and overlapping N_{\text{ov}} from Turn 2.	Quantifies the reconciliation and decision-regret error caused by artificial clipping and prefill-decode overlap double-counting.
A3: Wake-Up Blind (P_dec(1) -> slope)	Replaces solo wake-up power P_{\text{dec}}(1, M_{\text{solo}}) with incremental slope P_{\text{dec}}^+ - P_{\text{dec}}^0.	Isolates the exact Joule savings from consolidating arrivals onto active batches vs. waking idle GPUs (n_w = 0 \to 1).
A4: Prefix-Blind (p_hit = 0) & Static Decode Length	Sets p_{t,w}^{\text{hit}} \equiv 0 (A4a) or replaces survival \tilde{d}_{w,\text{rem}} with static t=0 \hat{d}_j (A4b).	Isolates the energy value of pricing verified KV prefix hits and dynamic survival updating on heavy-tailed outputs.
A5: Additivity Over Node-Level Phase DVFS	Enables *PowerSlider*/*Tri-serve* 810 MHz decode knee locking beneath B1, B4, B5, and	Tests whether cluster-level marginal-Joule dispatch remains additive when

Ablation ID	Exact Modification to PIGOU-J Final	Physical Mechanism Isolated
	PIGOU-J Final.	node-level DVFS is active.

6.3 Experiment-to-Claim Evidence Map

Claim ID	Exact Scientific Claim	Verification Vehicle	Primary Metric & Statistical Test	Current Epistemic Status
Claim 1	Ex-post Shapley attribution (\phi^{\text{Shapley}}) equals ex-ante counterfactual marginal energy (\Delta E_{\text{marginal}}) iff batch energy is strictly additive, and misorders dispatch under submodular wake-up/HBM amortization.	**Theorem 1** (Sec. 3.2) + **Exp 2** (M=100 deterministic 2-worker prefix replays).	Mean pairwise decision regret \bar{R}(\pi) in Joules; two-sided paired Wilcoxon test (p < 0.01).	**Mathematically Proved (Theorem 1); Hardware Exp 2 Pending**
Claim 2	Time-slice integration of the calibrated step-power surface reconciles with wall-clock NVML + RAPL + PDU energy within \pm 2.0\%.	**Protocol A** (Sec. 5.1) across **Exp 0, Exp 3, Exp 4**.	Relative reconciliation residual $	\epsilon_{\text{rec}} }
Claim 3	On homogeneous multi-replica pools, PIGOU-J Final reduces Total Node \text{J/1K goodput tok} by \ge 4.0\% over MARCS-SC and JCalib-Shapley under diurnal light load (\rho=0.30) and shared-prefix load (\rho=0.70).	**Exp 3** (N = N_{\text{req}}(\hat{\sigma}_{\text{diff}}) paired seeds per load regime).	Total Node \text{J/1K goodput tok}, P99 TTFT, P99 TPOT; Holm-Bonferroni Wilcoxon (\alpha_{\text{FWER}}=0.05) + 95% BCa CI.	**Unverified Hypothesis (Requires Exp 3 Execution)**
Claim 4	All-step ring-buffer IW-dACI bounds population unsafe	**Theorem 2** (Sec. 3.3) + **Exp 5** (20-seed 4-phase	Empirical population unsafe deferral rate &	**Mathematically Proved (Theorem 2); Exp 5 Pending**

Claim ID	Exact Scientific Claim	Verification Vehicle	Primary Metric & Statistical Test	Current Epistemic Status
	deferral \frac{1}{T-h}\sum \text{err}_t \le \alpha + O(1/T + 1/(p_{\text{audit}}\sqrt{T})) under non-stationary drift, whereas audit-only scaled updates fail.	drift replay).	rolling 200-query excess risk vs. \alpha.	
Claim 5	On two-tier heterogeneous pools, PIGOU-J Final + IW-dACI reduces Total Node \text{J/1K goodput tok} (including shadow-audit Joules) by \ge 7.0\% over Static Conformal + Token Cost at empirical risk \le \alpha.	**Exp 4** (N = 10 paired seeds on 4–8 GPUs).	Net Node \text{J/1K goodput tok} (audit energy included) & empirical risk on LiveBench/LiveCodeBench.	**Unverified Hypothesis (Requires Exp 4 Execution)**

7. Independent Hostile Review, Limitations, and Submission-Readiness Criteria (Requirement 5)

7.1 Final Adversarial Review of PIGOU-J Final

We subject PIGOU-J Final to a final hostile review across five dimensions, recording which objections are permanently resolved by our repairs and which remain empirical blockers requiring hardware data:

1. **Mathematical & Algorithmic Correctness Review:**
    - *Challenge:* "Does \Delta \hat{E}_{\text{self}} + \Delta \hat{E}_{\text{ext}} still violate energy conservation if P_{\text{dec}}(n_w+1, M_w^+) < P_{\text{dec}}(n_w, M_w), does chunked prefill still double-count overlap steps, and does IW-dACI still mis-scale updates by 1/p_{\text{audit}}?"
    - *Resolution Status:* **100% RESOLVED.** Removing max(0.0, ...) restores the exact algebraic identity P^+\tau^+ - P^0\tau^0 \equiv (P^+ - P^0)\tau^0 + P^+(\tau^+ - \tau^0) for all real values; Stage I and Stage II explicitly deduct K_{\text{wait}} and K_{\text{pre}} before computing N_{\text{ov}}; and IWDACIRingBuffer executes on every arrival t, pushing 0.0 on unaudited/large steps and \text{err}_t / p_{\text{audit}} on audited steps.
2. **Novelty & Positioning Review vs. MARCS-SC and JouleShare:**
    - *Challenge:* "Why isn't PIGOU-J Final a straightforward combination of MARCS-SC

(latency externality) and JouleShare (batch energy profiling)?"
    - *Resolution Status:* **RESOLVED CONCEPTUALLY; PENDING EMPIRICAL CONFIRMATION IN EXP 2 & EXP 3.** Theorem 1 proves mathematically that JouleShare's Shapley value \phi_i^{\text{Shapley}} is the wrong target for dispatch and reverses optimal worker ordering under submodular wake-up/HBM amortization. Meanwhile, against MARCS-SC, PIGOU-J Final exposes a fundamental physical conflict between latency-externality minimization (which spreads load onto idle n_w=0 GPUs to avoid incumbent delay) and Joule-externality minimization (which packs load onto active n_w \ge 1 GPUs to avoid the 150\text{–}200\text{ W} active wake-up jump and weights prefill KV hits by prefill wattage P_{\text{pre}} > P_{\text{dec}}).
3. **High-Load Homogeneous Saturation Limitation (Explicitly Preserved Blocker):**
    - *Challenge:* "When a homogeneous cluster operates at heavy steady saturation (\rho \ge 0.85) with zero prefix cache hits, every worker has n_w \ge 8 (no wake-up jumps) and GPU decode power sits on its flat HBM-saturated plateau (\sim 350\text{ W} on H100). In that regime, \Delta \hat{E}_{\text{marginal}} \approx P_{\text{plateau}} \cdot \Delta \hat{L}_{\text{social}}, so PIGOU-J Final will make nearly identical decisions to MARCS-SC."
    - *Resolution Status:* **PRESERVED AS AN EXPLICIT PHYSICAL LIMITATION.** We do not pretend PIGOU-J Final beats MARCS-SC in cache-miss heavy saturation on homogeneous GPUs; instead, our evaluation explicitly sweeps \rho \in \{0.25, 0.50, 0.70, 0.90\} to demonstrate the boundary where Joule-externality routing outperforms latency-externality routing and where the two converge.
4. **Shadow-Audit Energy Overhead in IW-dACI (Explicitly Preserved Blocker):**
    - *Challenge:* "In Theorem 2, i.i.d. Bernoulli(p_{\text{audit}}) auditing requires running p_{\text{audit}} of small-model traffic on m_{\text{large}} (which consumes 3\text{ to }4.4\times more energy per token), imposing a 15\text{ to }22\% energy tax at p_{\text{audit}} = 0.05, while small p_{\text{audit}} widens the Azuma-Hoeffding bound by 1/p_{\text{audit}}."
    - *Resolution Status:* **PRESERVED TRADE-OFF.** All shadow-audit GPU Joules are strictly included inside E_{\text{measured}} in Exp 4, and we sweep p_{\text{audit}} \in \{0.01, 0.02, 0.05, 0.10\} to characterize the exact Pareto frontier between audit energy tax and risk concentration width.

7.2 Reproducibility and Submission-Readiness Checklist

The research dossier is **100% mathematically verified and ready to enter Stage 0 of the physical pilot**. It may be upgraded from **READY FOR PILOT ONLY** to **READY FOR CONFERENCE SUBMISSION** *if and only if* all five physical verification gates below are completed and logged:

Gate ID	Mandatory Artifact / Measurement Required Before Submission	Acceptance Threshold	Current Status
Gate 1	**Stage 0 AA Repeatability & Time-Slice Reconciliation Log**	Measured \hat{\sigma}_{\text{diff}} \le 4.0\% and time-slice reconciliation error $	\epsilon_{\text{rec}}

Gate ID	Mandatory Artifact / Measurement Required Before Submission	Acceptance Threshold	Current Status
	(Exp 0)		
Gate 2	**Stage 1B Counterfactual Prefix-Replay Dataset (Exp 2)**	PIGOU-J Final achieves statistically significant lower pairwise Joule regret than both Token-Proportional and JCalib-Shapley across M=100 busy-period replays ($p < 0.01$, 3.3 hours).	**Unmeasured (Hardware Required)**
Gate 3	**Stage 2 Homogeneous 4-GPU Evaluation (Exp 3)**	Completed N = N_{\text{req}}(\hat{\sigma}_{\text{diff}}) paired-seed runs across \rho = 0.30 and \rho = 0.70, clearing Kill Criteria K3 and K4 against MARCS-SC and JCalib-Shapley (20.0h MVP / 45.8h full).	**Unmeasured (Hardware Required)**
Gate 4	**Stage 3A & 3B Non-Stationary IW-dACI & Two-Tier Evaluation (Exp 4 & Exp 5)**	Empirical unsafe deferral rate satisfies Theorem 2 bound across 20 drift seeds, and net two-tier node energy (including audit Joules) improves by \ge 7.0\% over Static Conformal + Token Cost (20.9 hours).	**Unmeasured (Hardware Required)**
Gate 5	**Open-Source Reproducibility Package**	Public repository containing pigou-profiler, IWDACIRingBuffer, pigou_j_final_dispatch, raw 50 ms NVML/RAPL/PDU CSV logs, seeded trace generators, and scipy.stats analysis scripts.	**Code Specified in Sec. 4.3; Repo Packaging Pending Runs**